Ecozones of Canada

Starting a Project

Importing a Map

Editing the Legend

Recoding Map Layers

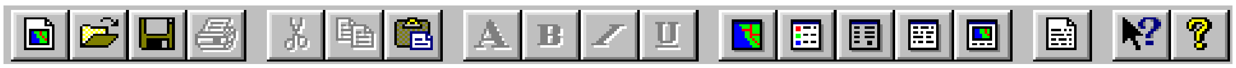
Adding Overlays to a Map

Georeferencing

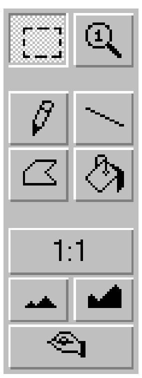
MFworks Lesson

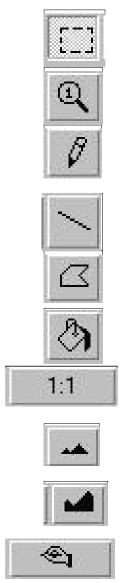
MFworks Icons and Tool Bars

Create new map	Open map	Save Map	Print Map		Cut	Copy	Paste		Font	Bold	Italic	Underline		View Legend	Select Legend	Display Information	Display Comment	Display Layout		New Script		Help	About



Tool Bar





Select Tools
Numeric Magnifying Glass
Freehand Drawing Tool
Draw Line Tool
Draw Polygon Tool
Fill Tool
Zoom To
Zoom Out
Zoom In
Redraw Map

For more information check http://www.thinkspace.com

GEOMATICS
FOR HIGH SCHOOL CURRICULUM

STARTING MFWORKS

Before you begin, the lesson data folder must be ***copied*** into a drive in which the student has writing and editing capabilities. This will probably be the drive containing your own personal account on the school's system or it may be the **C: drive** of the machine you are currently working on. If the latter is the case you will have to work on the same machine each day until this assignment is complete. Each school set-up is different so work with your teacher to prepare your data folder.

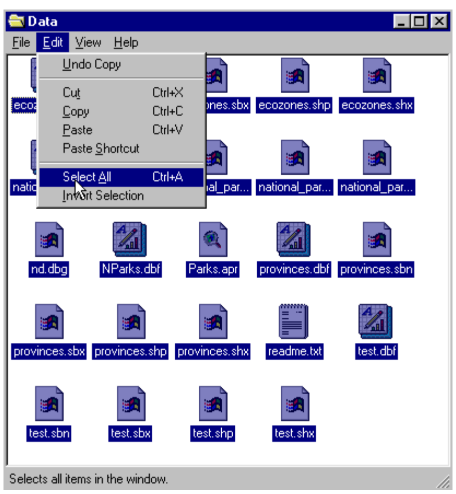
! *(The data for the lesson must first be downloaded from this Web site and placed on the school computer or network. Ask your teacher where to find the data folder that has been downloaded).*

To do this, use the file manager program on your computer. For windows based systems this will be **My Computer / Windows Explorer /.**

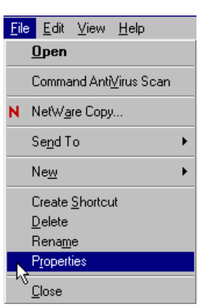
Once the data folder has been ***copied*** to the required destination the files must be made usable by removing the ***read only*** attribute. For windows based systems follow these instructions:

!*(The data downloaded from the Web Site is set as* ***read only*** *so that it can not be changed or altered by accident.)*

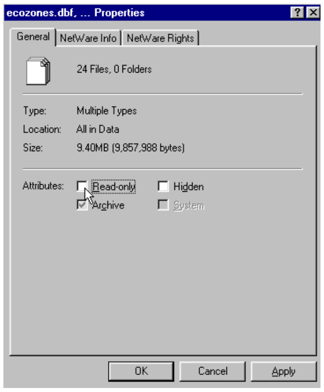
1. Open your new data folder
2. Choose **Edit >** ***Select all*** (they will all appear Blue)



3. From the menu choose:
   File > Properties



4. Remove the check mark in the box beside the label ***read only***.



5. Click **apply**, then **OK**

The data in your new data folder can now be opened, edited and saved using MF Works to carry out the following project.

! Denotes an observation.

? Denotes a question to be answered on the Student Activity Sheet.

Begin the lesson by starting up the **MFworks** program.

1. From your system Start menu, choose **Programs > MFworks > MFworks.**

The MFworks program will launch. If necessary, click the window maximize button so that the MFworks window fills the screen.

! *When the program starts, it automatically opens a window titled "Untitled Project." MFworks uses **projects** to group together and organize maps that are used together. A project has been created for you to use with this activity, so you do not need the default project.*

2. Choose **File > Close** to close the Untitled Project window.
3. Choose **File > Open Project** to bring up the Open dialog box.
4. Make sure that the folder named in the "Look in" field is the data folder just created.
5. Select the file name **Lesson1.mfp** and click the Open button to open that project.

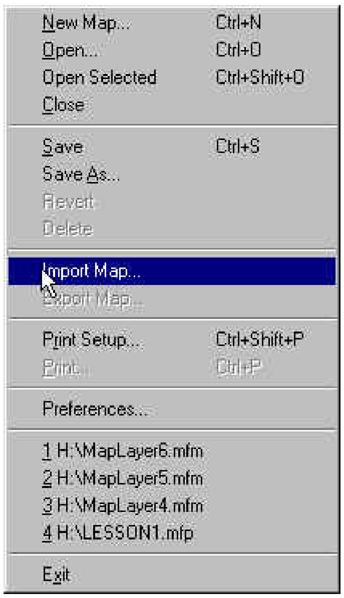
! *The **Lesson1** project lists three maps that you will use in this activity. You will also import a map and create several others using MFworks operations. As they are created, these new maps will automatically be listed in the Project window. In order to keep the workspace usable it is important to close or minimize any maps not in use leaving only one map open on the screen to work on. If you close a map ignore the "Save Changes" message where appropriate.*

IMPORTING A MAP SHOWING THE LANDCOVER OF CANADA

The data files for this activity include an image showing the land cover of Canada. The image is a TIFF (Tagged Image File Format) file. To use this information, the file must be imported into **MFworks.**

1. Click on **File > Import Map** … to open the Import dialog box.

Make sure that the folder named in the "Look in" field at the top of the dialog box is the data folder created for the student.



2. At the bottom of the dialog box, use the drop down list to enter **TIFF** (Tagged Image File Form) in the "Files of Type" field.

3. Select the file name AVHRR.TIF and click the **Open** button to import the image.

! *The map shows the land cover of Canada, designated by colour. It also shows latitude and longitude lines. It has been given the default name "**MapLayer1**."*

4. Check the Project window. The name "MapLayer1" appears there. The icon beside the name is dimmed because the map has not yet been saved.

5. Click and drag the edge or corner of the "**MapLayer1**" window to enlarge the Map window to fill your screen.

! *The map can be viewed at different levels of magnification, called **zoom levels**. The map has opened at a default zoom level of 1:1. shown on the Zoom Ratio button on the left side of the Map window.*

At this point, try the zoom tools.

6. Click on the **zoom in** symbol tool. To view the map at a higher zoom level. Notice that the **Zoom Ratio** button now reads 1:2.

7. Click on the **zoom Ratio** button to display the Zoom Ratio menu. Choose 1:16.

? Do you see the pixel structure of the map?

! *The map is made up of many small square cells or pixels. Each cell represents a square parcel of land. In the map, each cell has a colour and a numeric value. All of the cells having the same value together form a **map zone**.*

You can view the numeric values using the Numeric Magnifying Glass tool.

8. Change the zoom level of the map back to 1:1, either by clicking the Zoom out button repeatedly, or by displaying the Zoom Ratio menu and choosing 1:1.

9. Click on the **numeric magnifying glass symbol** button on the left side of the Map window.

10. Move the cursor over the map. Notice that the cursor appear as a small magnifying glass.

11. Choose an area of the map that you are interested in, and click the mouse button. A 5-by-5 matrix of values will appear below the magnifying glass. The matrix shows the values of the map cells (pixels) currently below the magnifying glass.

12. Experiment with finding the values associated with different colours in the map.

EDIT THE LEGEND

! *All MFworks maps have an associated **legend** that lists all of the map zones. The legend can be used to display information about the map zones and to manipulate the colours used in the map.*

We want to change the legend that identifies the land cover classes by number to one with land cover class names.

1. Resize the Map window so that it takes up about half of the space available on the screen.

2. If you cannot see the Legend window, click the **Legend** button to make the Legend window the active window.

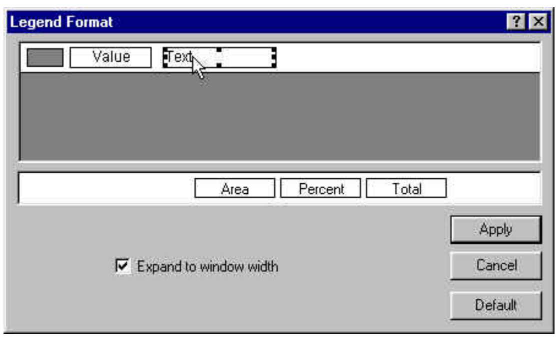
3. If necessary, click and drag the Legend window so that it is visible beside the Map window.

! *The Legend window displays a column of colour elements, showing the colours used in the map, and two columns of numbers. The first column of numbers gives the cell values (the values that you viewed using the Numeric Magnifying Glass) of the map zones. The second column* *gives the number of cells in each map zone. The Legend window is in the default format. You can change the format to something more useful*

4. From the Main Menu bar choose **Legend > Format** to open the Legend Format dialog box.

! *The different types of information that can be displayed in the legend are represented by small, rectangular elements. The elements that you want to have shown in the legend are placed and positioned in the Legend Entry Prototype area at the top of the dialog box. The elements that you do not wish to display are stored in the Legend Element Holding area in the lower part of the dialog box.*

You will use the Legend Format dialog box to create a legend that shows map zone colours, values, and descriptive text.



5. Click on the **Area** element and drag it to the Legend Element Holding area.
6. Click on the **Text** element and drag it to the left so that it is right beside the Value element in the Legend Entry Prototype area.
7. Click and drag the lower right element handle (the small black square in the lower right corner of the element) and enlarge the text element to approximately twice its original size.)
8. Click the **Apply** button.

! *The Legend window will now show colour elements and one column of numbers - the cell values. There are blank spaces available for text, but you must enter the text.*

9. Click beside the map value "0." A flashing cursor will appear.
10. Type the word **"urban."** The legend now tells you that the map zone that is coloured black has the value "0" and represents urban areas.
11. Type legend text for the rest of the map zones, using the information shown below:

Value	Legend text	Value	Legend text
0	Urban	**175**	Shrubland & alpine
12	Coniferous forest	**176**	Mixed deciduous
30	Taiga	**219**	Barrens
31	Snow & ice	**224**	Forest burns
90	Mixed coniferous	**248**	Grassland
146	Tundra	**252**	Agriculture
168	Deciduous forest	**255**	Water

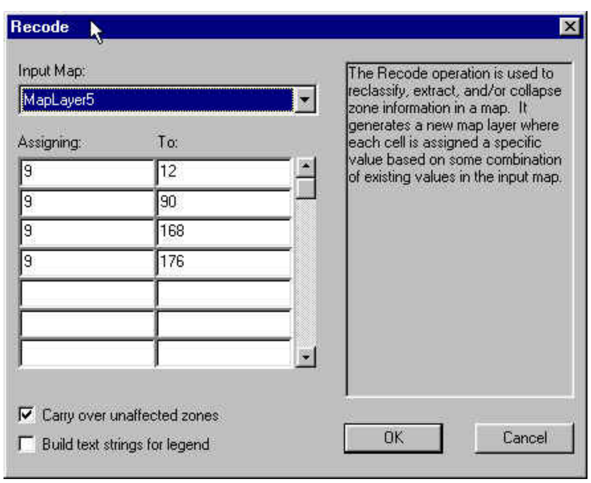
GROUPING TOGETHER MAP ZONES USING THE RECODE OPERATION

! *The map shows five zones representing forest: coniferous forest, mixed coniferous, deciduous forest, mixed deciduous, and forest burns. You can use the Recode operation to create a new map that groups these five zones together into a single zone that just represents "forest."*

1. Use the legend to fill in the values for the five forest map zones:

Value	Zone
	Coniferous forest
	Mixed coniferous
	Deciduous forest
	Mixed deciduous
	Forest burn

2. In the Main Menu bar choose **Operations > Recode** to open the Recode dialog box. This dialog box allows you to use an operation that creates a new map by assigning different values to the map zones of an existing map.
3. Use the drop-down list to enter the map name MapLayer1 in the "Input Map" field.
4. You will assign the value "20" to all map zones representing forest. To do this, click in the first "Assigning" field and type the value 20, and then click in the first "To" field and type the value for coniferous forest.
5. Continue the process by typing "20" in the "Assigning" field and the current value in the "To" field for each of the remaining forest types.
6. If necessary click in the box labelled "Carry over unaffected zones" so that a check mark appears there.
7. Click the **OK** button to execute the operation.



! *The operation will execute and a new map, named* ***MapLayer2.*** *It will open and be listed in the Project window.*

8. Position the Map and Legend windows on the screen so that you can see both of them.
9. Click on the Legend window to make it the active window, then click beside the value "20" and type the text "forest."

! *You can use the Legend window to change the colour of the map zone representing forest.*

10. Click on the colour element for the map zone representing forest. The Colour Palette will open.
11. Click on a dark green square to assign that colour to the map zone and to close the Colour Palette.
12. Use the Numeric Magnifying Glass tool to look at the values of the dark green cells.

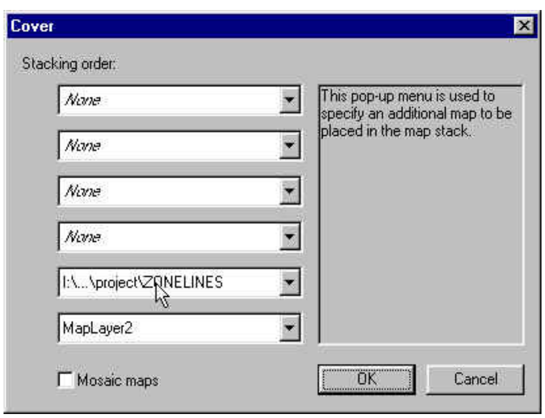
ADDING OVERLAYS USING THE COVER OPERATION

! *Now we want to add the ecozone boundary lines to the land cover raster image in order to investigate the land cover characteristics of each ecozone. This is carried out by overlaying an ecozone boundary line map layer on the land cover map layer. The overlaying of map layers is one of the basic operations of all GIS systems and analysis.*

1. Choose **Operation > Cover** to open the cover operation window. The cover operation generates a new map layer through stacking several map layers on top of each other. Each new cell value is the top most non-VOID cell that appears in the map stack. VOID cells are treated as transparent. The operation looks down through the top of the stack to record all visible cells.

! *This will bring up a stacking order menu. Because we want the ecozone lines to be on top of the land cover map, we will choose the map (MapLayer2) as the bottom layer and "zonelines" as the next layer up.*

2. Click **O.K.** to produce a new map "*MapLayer3"* showing the overlay of the two maps.
3. Click and drag the edge or corner of the "*MapLayer3*" window to enlarge the Map window to an appropriate size for viewing. Study the land cover make-up of the different Ecozones as defined by the red boundary lines. Select one of the Ecozones and write a short description of the land cove. (Optional)



GEOREFERENCING YOUR MAP

! *Ever position on the earth's surface can be given a geographic position so that its exact location can be found again and again. This is done by assigning latitude and longitude lines to the globe. The globe is divided into 360 degrees of longitude beginning at the Greenwich Mean line. This line runs north / south across England (near London) and across the eastern portion of Africa. To* *facilitate readings for GIS positioning, all lines up to 180 degrees east of the Greenwich Mean line are given a positive value. All longitudes up to 180 degrees west are given a negative value. Canada lies mainly between –55 and –145 (55 West to 145 West) degrees. In the same fashion, the globe is divided into 180 degrees of latitude. The 90 degrees north from the equator are positive, the 90 degrees south from the equator are negative. Canada lies mainly between 42 and 86 (42 North to 86 North) degrees. There are a number of other georeferencing systems that can be used to identify position. These are based on maps and map projections.*

At present our map is given coordinates (rows and columns) based on map cells (pixels). This exercise will change the rows and columns of cells in our map to latitude and longitude that represent Canada's position on the globe.

1. First, Make sure that your tracker is selected.
2. Choose ***Map*** >**Tracker** and make sure there is a check beside it. The tracker displays the position of the pointer within the geometry of the map layer. The tracker also displays position in the coordinate system specified by the Ruler Coordinates command.

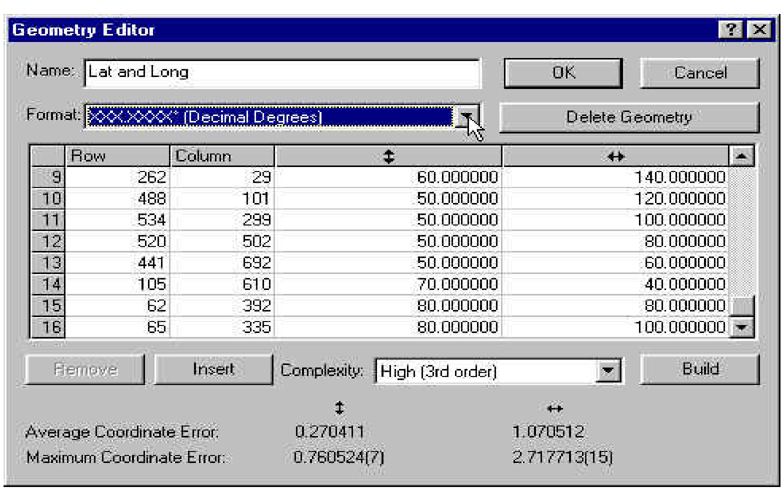
! *Make sure that the bottom of you screen appears as below with numbers in the rows and columns boxes. Notice how the numbers change as you move the cursor over the map.*

3. Place your cursor on the intersection of a Latitude line and Longitude line on your map.
4. Note the row and column values, and the latitude and longitude values for the intersection. You can find the row and column values by looking on tracker bar (shown above).
5. Find and write down 16 sets of these coordinates. These are often referred to as control points or ground control points because we know both the map position and ground position for each point. These points can be used in a mathematical equation to calculate and assign ground position to every cell in the map.

Use the table on the activity page to help you.

6. Choose **Map > Edit geometry**/ **new.** To open a table in which the recorded values can be entered.

7. Click **insert**. Enter all 16 sets of coordinate points for row, column, latitude, and longitude (which you previously wrote down).
8. Select **complexity high (3rd order).**
9. Click **build.**
10. Click **O.K.** to build a new geometry for the map that georeferences it to latitude and longitude.



! Note that because our map covers such a large area and is located in the north where longitude lines converge our new geometry is only approximate and therefore position and distance measurements are only approximate.

LOCATING PLACES AND MEASURING DISTANCES ON YOUR MAP

! We are now able to find the latitude and longitude of any location on our map. It is also possible to measure distances between places.

1. Choose **Map > Ruler Coordinates/ Geometry.** Rulers provide a scale for calculating distances and determining the coordinates of cells in the Map window. Arrows track the position of the pointer on each scale.

! Your tracker window should look like this now.
Notice now that it tells you the latitude, longitude and the distance between two points.

2. Move your cursor over Toronto (black area just above Lake Ontario remember black is the colour for urban areas).

? What are its latitude and longitude coordinates? Also find Winnipeg, Yellowknife, Vancouver, and Halifax?

? Measure the distance between Vancouver (British Columbia) and St. John's (Newfoundland).

GROUPING AND RECOLOURING ECOZONES

! Just as you regrouped forest land cover in the earlier part of the lesson, you can group ecozones. However this time we will recolour the legend.
First, close the map layers that are currently open by clicking on the close symbol button in the right hand corner.

1. From your project window, double click on the "*ecozones"* map.

Now you are going to group by recolouring all of the Cordillera Ecozones.

2. Click and drag the corner of the ecozone legend window to enlarge it to a workable size. If you can't find the legend window click on the legend button to bring it to the front and make it active. (Note that the MFWorks legend is in a different order than the legend on the map. Please use the MFworks legend for this exercise and disregard the map legend).
3. Click on the first Cordillera ecozone (Boreal Cordillera) to select and put a box around it.
4. Click on the colour box to activate the colour palette.
5. Select and click on a new colour from the palette that will represent Cordillera ecozones. (Remember this new colour so that you can assign it to the other Cordillera ecozones).
6. Click and select the next Cordillera ecozone (Arctic Cordillera) and recolour it the same as in step 5. Repeat for all the Cordillera ecozones.

? What can you say about the distribution of Cordillera ecozones? What do you think the word cordillera means?

Now recolour and group the following together.

1) The shields ecozones
2) The arctic ecozones
3) The plains ecozones

? Can you see a pattern?

7. Now repeat the recolouring process for the Boreal, and Taiga ecozones.

* Remember to use colours that have not already been used.